Chapter 2

Minimization Using Exponentials

We recall the combinatorial discrepancy problem for set systems:

> Let X be a finite set, and let $\mathcal{R}$ be a set of subsets of X. We call $(X, \mathcal{R})$ a *set system* with *vertex set* X.
>
> Given $(X, \mathcal{R})$, a *two-coloring* of X is an assignment of a -1 or a $+1$ color to each element of X. Let χ denote a fixed two-coloring of X. That is,
>
> $$\chi \colon X \to \{-1, +1\}\,.$$
>
> Then, the *discrepancy* of a set $S \subseteq X$ with respect to χ is defined as
>
> $$\operatorname{disc}_\chi(S) = \left| \sum_{v \in S} \chi(v) \right|.$$
>
> Define the *discrepancy* of $\mathcal{R}$ with respect to χ as
>
> $$\operatorname{disc}_\chi(\mathcal{R}) = \max_{S \in \mathcal{R}} \operatorname{disc}_\chi(S).$$

The algorithm of Theorem 1.4 (page 16) constructed a coloring with discrepancy $O\left(\sqrt{n \ln m}\right)$, by coloring the elements of X iteratively, as follows.

> Assume we have already colored $v_1, \ldots, v_{t-1}$ in earlier iterations. We next set the color of v_t to one that will minimize the sum
>
> $$\sum_{S \in \mathcal{R}} \Big(\operatorname{disc}^{t+1}(S) \Big)^c.$$
>
> Here $\operatorname{disc}^t(S)$ refers to the discrepancy of S at the start of iteration t (after having colored $v_1, \ldots, v_{t-1}$).

In this chapter, we introduce the following powerful new idea, which at first seems purely technical:

we run the previous iterative algorithm, except we instead minimize the function

$$\sum_{S \in \mathcal{R}} \exp\left(\eta \operatorname{disc}(S)\right),$$

where we carefully choose the parameter $\eta > 0$.

The best value of η will follow from the technical calculations; but one can already guess it by comparing it to the previous function and the value of the parameter c there.

Recall that we had set $c \approx \ln m$ in Theorem 1.4 (page 16).

As $(\operatorname{disc}(S))^c$ can be written as $\exp\left(c \ln \operatorname{disc}(\boldsymbol{S})\right)$, we compare this form with our earlier polynomial function, assuming $\operatorname{disc}(S) = \Theta\left(\sqrt{n \ln m}\right)$ for each S:

$$\textbf{Earlier function:} \qquad \sum_{S \in \mathcal{R}} \left(\operatorname{disc}(S)\right)^c \approx \sum_{S \in \mathcal{R}} \exp\left(\ln m \cdot \ln \sqrt{n \ln m}\right).$$

$$\textbf{New function:} \qquad \sum_{S \in \mathcal{R}} \exp\left(\eta \operatorname{disc}(S)\right) \approx \sum_{S \in \mathcal{R}} \exp\left(\eta \cdot \sqrt{n \ln m}\right).$$

This indicates the right ballpark for the value of η—as we will see, we'll set $\eta = \Theta\left(\sqrt{\frac{\ln m}{n}}\right)$.

Moving to the exponential function simplifies the technical calculations—products of exponentials become sums!

Remark: Readers who have taken a basic course in probability theory are already familiar with this trick—it is called the *moment-generating function*.

In a sense, the function $\exp(x)$ can be considered as optimizing all the polynomials ("moments") of x *simultaneously*. The parameter η determines which polynomials of x is given the most weight.

Most concentration bounds—Hoeffding, Azuma-Hoeffding,...—are proved using this trick, though a polynomial function would work there as well (see Exercises 1.2.3 and 1.1.6 (pages 12 and 24)).

This phenomenon—that is, that one can achieve bounds derived from the exponential function also using polynomial functions—is a common theme in combinatorics (we leave a fuller discussion to Section 7.2 (page 178)). We illustrate this with the following simple example. Assume throughout that n is an even integer.

Claim 2.1. *Let $G = (V, E)$ be a graph on n vertices and m edges.*

*There exists a partition of V into two **equal sized** subsets, say U and $V \setminus U$, such that the number of edges of E with exactly one endpoint in U is at least*

$$\frac{m}{2} \cdot \frac{n}{n-1}.$$

Proof. Let U be a subset of V of size $n/2$, chosen uniformly at random from among all $\binom{n}{n/2}$ such subsets. Then for any edge $\{v_i, v_j\} \in E$, the probability that exactly one of

$\{v_i, v_j\}$ is picked in U is precisely

$$2\,\frac{\binom{n-2}{n/2-1}}{\binom{n}{n/2}} = 2\,\frac{(n-2)!}{(\frac{n}{2}-1)!(\frac{n}{2}-1)!}\cdot\frac{(\frac{n}{2})!(\frac{n}{2})!}{n!} = 2\frac{\frac{n}{2}\frac{n}{2}}{n(n-1)} = \frac{1}{2}\cdot\frac{n}{n-1}.$$

By linearity of expectation, the expected number of edges with exactly one endpoint in U is precisely $\frac{m}{2}\cdot\frac{n}{n-1}$. Thus, such a U must exist. □

The proof above constructs a U with size *precisely* $n/2$—this makes sense intuitively, as one would expect an equi-partition to maximize the number of edges crossing that cut. If we are willing to lose a slight factor, constructing U uniformly at random from all subsets of V gives a simpler proof.

Claim 2.2. *Let $G = (V, E)$ be a graph on n vertices and m edges. There exists a partition of V into two subsets, say U and $V \setminus U$, such that the number of edges of E with exactly one endpoint in U is at least*

$$\frac{m}{2}.$$

Proof. Construct U by picking each vertex of V into U independently with probability $\frac{1}{2}$. Then for a fixed edge $\{v_i, v_j\} \in E$, the probability that exactly one of $\{v_i, v_j\}$ is picked in U is $1/2$. By linearity of expectation, the expected number of edges with exactly one endpoint in U is precisely $\frac{m}{2}$. □

Essentially, the second proof is easier because averaging results in *implicit cancellation* of terms. The cancellations can be seen in the following explicit calculation:

$$\Pr\left[\text{exactly one of } \{v_i, v_j\} \text{ is picked}\right] = \sum_{s=1}^{n-1}\frac{2\binom{n-2}{s-1}}{2^n} = \frac{2}{2^n}\sum_{t=0}^{n-2}\binom{n-2}{t} = \frac{2}{2^n}\cdot 2^{n-2} = \frac{1}{2}.$$

Above, we see that we do not have to evaluate each binomial term; the sum can be nicely calculated without it! We should note that $|U|$ is sharply concentrated around $n/2$—in fact, with high probability, the size of U will be in the interval $\frac{n}{2} \pm O(\sqrt{n})$.

In this chapter we will use the above exponential weight function to re-derive the bound of Theorem 1.4 (page 16). The structure of the proof will be analogous, but the calculations will be shorter.

Remark: As in the above partitioning example, the exponential weight function does not give tighter bounds, but rather is easier to manipulate than the polynomial function. See Exercise 1.2.3 (page 24).

The second section of this chapter shows another application of exponential minimization, the construction of small ϵ-approximations. In this application, we will also encounter a variant of exponential minimization: instead of the exponential weight

$$\exp\left(\eta\sum_{i=1}^{T} X_i\right),$$

$\eta \le 1$ and $X_i \in [-1, 1]$, which results from a multiplicative weight update, we can also use the first

two terms of the Taylor expansion of $\exp(\cdot)$ in order to obtain the weights

$$\prod_{i=1}^{t}(1+\eta X_i)$$

that come from a multiplicative update by the factor $(1+\eta X_i)$ in each iteration i.
As we will see later, in addition to simpler calculations, it can lead to better bounds as well.

2.1 Discrepancy, II (algorithm for $\sqrt{n \ln m}$ using $\exp$)

> *When lecturing to a student body different from the one at your own institution there is an irresistible temptation to cut corners, omit difficult details, and experiment with teaching methods. Any wounds to the students' development caused by the peculiar point of view will be left behind as someone else's responsibility to heal.*
>
> Richard Feynman

The main result of this section is a second proof of the following theorem—in fact, we give two different proofs.

Theorem 1.4. *Given a finite set system $(X, \mathcal{R})$ with $m = |\mathcal{R}|, n = |X|$ such that $m \geq n \geq 2$, there is a greedy algorithm that colors the elements of X in the order $v_1, \ldots, v_n$ such that the final coloring $\chi \colon X \to \{-1, +1\}$ satisfies*

$$\text{for all } S \in \mathcal{R}\colon \quad \operatorname{disc}_\chi(S) = O\left(\sqrt{n \ln m}\right).$$

Here

$$\operatorname{disc}_\chi(S) = \left|\sum_{v \in S} \chi(v)\right|.$$

Further, the algorithm assigns a color to v_t in iteration t using only the information on elements $v_1, \ldots, v_t$.

Overview of ideas. The new idea is to maintain set weights that depend *exponentially* on the current discrepancy of each set.

The key point is that, as the weight of a set grows exponentially with its discrepancy, minimizing the total weight will, approximately, minimize the maximum discrepancy of any set of $\mathcal{R}$.

Define the weight of a set $S \in \mathcal{R}$ at the start of iteration t as:

$$y^t(S) = \exp\left(\eta \cdot \operatorname{disc}^t(S)\right),$$

where $\operatorname{disc}^t(S)$ denotes the discrepancy of S at the start of iteration t, with respect to the colors of $\{v_1, \ldots, v_{t-1}\}$.

Here $\eta > 0$ is a scaling parameter to be set later.

Thus, when coloring element v_t, we will assign it a color that minimizes the total weight

$$\Omega^{t+1} = \sum_{S \in \mathcal{R}} y^{t+1}(S).$$

We use the following insight:

for each $S \in \mathcal{R}$ with $\operatorname{disc}^t(S) > 0$ that contains v_t, one setting of v_t's color increases its discrepancy, and the other setting decreases it.

Thus, if all $S \in \mathcal{R}$ had non-zero discrepancy, we could use an averaging argument over the two color possibilities to show the following upper-bound on the weight increase:

$$\sum_{S\in\mathcal{R}} y^{t+1}(S) \leq \sum_{S\in\mathcal{R}} \left(\frac{1}{2} \cdot e^{\eta\left(\operatorname{disc}^t(S)+1\right)} + \frac{1}{2} \cdot e^{\eta\left(\operatorname{disc}^t(S)-1\right)} \right) = \left(\frac{e^{\eta}+e^{-\eta}}{2} \right) \sum_{S\in\mathcal{R}} y^t(S). \tag{2.3}$$

We will set η small enough so that we can use $e^z \leq 1 + z + z^2$ (Equation (A.8) (page 768)), to get

$$\frac{e^{\eta}+e^{-\eta}}{2} \leq \frac{\left(1+\eta+\eta^2\right)+\left(1-\eta+\eta^2\right)}{2} = 1+\eta^2.$$

Putting this into (2.3), and unrolling the recurrence, the total weight after all n iterations is

$$\sum_{S\in\mathcal{R}} y^{n+1}(S) \leq m \cdot \left(1+\eta^2\right)^n \leq m \cdot e^{n\eta^2},$$

since $1 + x \leq e^x$, for all $x \in \mathbb{R}$.

This gives the required bound: for every $S \in \mathcal{R}$, we have

$$\exp\left(\eta \cdot \operatorname{disc}(S)\right) \leq \mathbf{\Omega^{n+1}} \leq m \cdot e^{n\eta^2} \quad \Longrightarrow \quad \operatorname{disc}(S) \leq \frac{\ln m}{\eta} + n\eta = O\left(\sqrt{n \ln m}\right),$$

by setting $\eta = \sqrt{\ln m/n}$.

The above proof sketch does not *quite* work, for a somewhat annoying technical reason:

> The key property in Equation (2.3) (page 34) is that the discrepancy of each $S \in \mathcal{R}$ can *both* increase or decrease by 1. This is what allows us to upper bound the *average* multiplicative increase in the weight of each S by $\left(e^{\eta}+e^{-\eta}\right)/2$.
>
> However, this is not true when $\operatorname{disc}(S) = 0$—then the discrepancy of S can only increase by 1, no matter what color is given to v_t, and so the multiplicative factor becomes e^{η} instead of $\left(e^{\eta}+e^{-\eta}\right)/2$.
>
> The factor e^{η} is too big—it leads to the total weight of $me^{\eta n}$ at the end, which then gives a trivial bound of $O(n)$ for discrepancy.
>
> > This also exposes a subtlety in the earlier calculation that we glossed over: the factor η^2 instead of η in the exponent of the upper bound is the key to the whole proof. It essentially comes due to the fact that the linear terms in the Taylor expansion of $\exp(\eta \cdots)$ and of $\exp(-\eta \cdots)$ cancel out.

We present two ways around this problem:

Careful calculation. The point is that the weight increase for S is small when $\operatorname{disc}^t(S) = 0$—it increases from $e^{\eta \cdot 0} = 1$, to e^{η}. This increase is small enough to be incorporated in the calculations without significantly changing the upper bound.

Thus we minimize the same weight function—$\exp\left(\eta \operatorname{disc}(S)\right)$—at each iteration. But in upper bounding the weight increase from iteration t to $t+1$, we treat the sets of 0 discrepancy separately.

Different weight function. Our goal is to minimize $\operatorname{disc}(S)$—in other words, for each $S \in \mathcal{R}$, the number of '$+1$' colors should not be too large, and neither should the number of '-1' colors.

The weight function, $\exp(\eta \operatorname{disc}(S))$, captures this succinctly using the absolute value function inside $\operatorname{disc}(\cdot)$. But this has the drawback of making it insensitive to the case when $\operatorname{disc}(S) = 0$.

We can fix this by *separately* adding the two constraints for each S—one penalizing an excess of '$+1$' colors, and the other penalizing an excess of '-1' colors.

For each $S \in \mathcal{R}$, let P_S be the number of elements of color '$+1$', and N_S the number of elements of color '-1' (as before, the uncolored elements are considered to have the color 0). Then we minimize the weight function

$$\exp\left(\eta\left(P_S - N_S\right)\right) + \exp\left(\eta(N_S - P_S)\right) = \exp\left(\eta \cdot \operatorname{disc}(S)\right) + \exp\left(-\eta \cdot \operatorname{disc}(S)\right).$$

This symmetric weight now gives the desired multiplicative increase $\frac{e^{\eta}+e^{-\eta}}{2}$ for each set, including *sets with discrepancy* 0.



Proof of Theorem 1.4 (page 16) via 'careful calculation'. Let $\operatorname{disc}^t(S)$ be the discrepancy of $S \in \mathcal{R}$ at the start of the t-th iteration, after having colored the elements $v_1, \ldots, v_{t-1}$. Define weight

$$y^t(S) = \exp\left(\eta \cdot \operatorname{disc}^t(S)\right),$$

where $\eta > 0$ is a scaling parameter to be fixed later.

Let Ω^t be the total weight of sets of $\mathcal{R}$ at the start of iteration t:

$$\Omega^t = \sum_{S \in \mathcal{R}} y^t(S).$$

Note that for all $S \in \mathcal{R}$,

$$\operatorname{disc}^1(S) = 0 \qquad \text{and thus} \quad y^1(S) = \exp(\eta \cdot 0) = 1 \quad \text{and} \quad \Omega^1 = m.$$

Here is the algorithm, where $\eta > 0$ will be set optimally later.

Exp Discrepancy Algorithm 1 $\left(X = \{v_1, \ldots, v_n\}, \mathcal{R}\right)$

For $t = 1, \ldots, n$:

1. Assign v_t a color that realizes the following minimum:

$$\min\left\{\sum_{S \in \mathcal{R}} y^{t+1}\left(S \mid v_t \text{ is given color '+1'}\right), \sum_{S \in \mathcal{R}} y^{t+1}\left(S \mid v_t \text{ is given color '-1'}\right)\right\}.$$

   If both colors give the same total weight, choose either one arbitrarily.

2. Update the weights: for each $S \in \mathcal{R}$, set

$$y^{t+1}(S) = \exp\left(\eta \cdot \operatorname{disc}^{t+1}(S)\right).$$

Here is the key claim that states the existence of a coloring for v_t that does not increase the weights too much.

Claim 2.4. *At the start of the t-th iteration, we can assign a color to v_t such that*

$$\Omega^{t+1} = \sum_{S \in \mathcal{R}} y^{t+1}(S) \leq me^{\eta} + \left(\frac{e^{\eta} + e^{-\eta}}{2}\right) \cdot \Omega^{t}.$$

Proof. For the sets that do not contain v_t, the weight remains unchanged. Fix a set $S \in \mathcal{R}$ that contains v_t. There are two possibilities:

Case 1: $\operatorname{disc}^t(S) = 0$. In this case, irrespective of the color given to v_t, we have

$$\operatorname{disc}^{t+1}(S) = 1 \qquad \Longrightarrow \qquad y^{t+1}(S) = \exp\left(\eta \operatorname{disc}^{t+1}(S)\right) = e^{\eta}.$$

Case 2: $\operatorname{disc}^t(S) > 0$. Here, one of the color assignments for v_t must increase the discrepancy, and the other must decrease it. Thus, we can compute the average weight of S after coloring v_t, over the two possible color choices.

$$\frac{1}{2}e^{\eta\left(\operatorname{disc}^t(S)+1\right)} + \frac{1}{2}e^{\eta\left(\operatorname{disc}^t(S)-1\right)} = e^{\eta \operatorname{disc}^t(S)}\left(\frac{e^{\eta} + e^{-\eta}}{2}\right).$$

Putting the above two cases together, the average value of Ω^{t+1} over the two color choices can be upper bounded by

$$\sum_{\substack{S \in \mathcal{R}: \\ \operatorname{disc}^t(S)=0}} e^{\eta} + \sum_{\substack{S \in \mathcal{R}: \\ \operatorname{disc}^t(S)>0}} e^{\eta \operatorname{disc}^t(S)}\left(\frac{e^{\eta} + e^{-\eta}}{2}\right) \leq me^{\eta} + \left(\frac{e^{\eta} + e^{-\eta}}{2}\right)\Omega^{t}.$$

As the minimum is at most the average, this completes the proof of the claim. □

Applying Claim 2.4 (page 36) inductively and using $\Omega^1 = m$, we get

$$\begin{aligned}
\Omega^{n+1} &\leq me^{\eta} + \left(\frac{e^{\eta} + e^{-\eta}}{2}\right) \cdot \Omega^{n} \leq me^{\eta} + \left(\frac{e^{\eta} + e^{-\eta}}{2}\right) \cdot \left(me^{\eta} + \left(\frac{e^{\eta} + e^{-\eta}}{2}\right)\Omega^{n-1}\right) \\
&\leq \cdots \leq \sum_{t=0}^{n} \left(\frac{e^{\eta} + e^{-\eta}}{2}\right)^{t} \cdot me^{\eta} \leq (n+1) \cdot \left(\frac{e^{\eta} + e^{-\eta}}{2}\right)^{n} \cdot me^{\eta} \\
&\leq 2\, m^2\, e^{\eta} \left(\frac{e^{\eta} + e^{-\eta}}{2}\right)^{n},
\end{aligned}$$

because we assume that $m \geq n \geq 1$, and hence $2m \geq n+1$. Combining the upper and lower bounds on the total weight, we get

$$\max_{S \in \mathcal{R}} y^{n+1}(S) \;\leq\; \boldsymbol{\Omega^{n+1}} \;\leq\; 2m^2\, e^{\eta} \left(\frac{e^{\eta} + e^{-\eta}}{2}\right)^{n}.$$

Using the inequality $e^{\eta} + e^{-\eta} \leq 2e^{\eta^2/2}$ (Fact A.22 (page 770)) and substituting for $y^{n+1}(\cdot)$,

$$\exp\left(\eta \cdot \max_{S \in \mathcal{R}} \operatorname{disc}^{n+1}(S)\right) \;\leq\; 2m^2\, e^{\eta}\, e^{n\eta^2/2}.$$

Taking logarithms and dividing by η,

$$\max_{S\in\mathcal{R}} \operatorname{disc}^{n+1}(S) = O\left(\frac{\ln m}{\eta} + n\eta\right).$$

Setting $\eta = \Theta\left(\sqrt{\frac{\ln m}{n}}\right)$ to balance the two terms on the R.H.S. above, gives

$$\max_{S\in\mathcal{R}} \operatorname{disc}^{n+1}(S) \leq O\left(\sqrt{n\ln m}\right).$$

□

Remark: We can rephrase the proof of Claim 2.4 (page 36) succinctly in probabilistic language.

Imagine setting the color of v_t to '$+1$' or '-1' with equal probability. For any $S \in \mathcal{R}$ with $\operatorname{disc}^t(S) > 0$,

$$\mathrm{E}\left[y^{t+1}(S)\right] \leq \frac{1}{2}\cdot e^{\eta\left(\operatorname{disc}^t(S)+1\right)} + \frac{1}{2}\cdot e^{\eta\left(\operatorname{disc}^t(S)-1\right)} = e^{\eta\operatorname{disc}^t(S)}\cdot\left(\frac{e^{\eta}+e^{-\eta}}{2}\right).$$

Since the algorithm chooses the color minimizing Ω^{t+1}, the minimum is at most the average over the two color choices; linearity of expectation then gives

$$\begin{aligned}
\Omega^{t+1} \leq \mathrm{E}\left[\Omega^{t+1}\right] &= \mathrm{E}\left[\sum_{S\in\mathcal{R}} e^{\eta\cdot\operatorname{disc}^{t+1}(S)}\right] = \sum_{S\in\mathcal{R}} \mathrm{E}\left[e^{\eta\cdot\operatorname{disc}^{t+1}(S)}\right] \\
&= \sum_{\substack{S\in\mathcal{R}:\\ \operatorname{disc}^t(S)=0}} \mathrm{E}\left[e^{\eta\cdot\operatorname{disc}^{t+1}(S)}\right] + \sum_{\substack{S\in\mathcal{R}:\\ \operatorname{disc}^t(S)>0}} \mathrm{E}\left[e^{\eta\cdot\operatorname{disc}^{t+1}(S)}\right] \\
&\leq \sum_{\substack{S\in\mathcal{R}:\\ \operatorname{disc}^t(S)=0}} e^{\eta} + \sum_{\substack{S\in\mathcal{R}:\\ \operatorname{disc}^t(S)>0}} e^{\eta\cdot\operatorname{disc}^t(S)}\cdot\left(\frac{e^{\eta}+e^{-\eta}}{2}\right). \\
&\leq me^{\eta} + \left(\frac{e^{\eta}+e^{-\eta}}{2}\right)\cdot\Omega^t.
\end{aligned}$$

From here, we can continue the proof as above.

Remark: Note that the statement and proof of Claim 2.4 (page 36) is analogous to the statement and proof of Claim 1.5 (page 19) in Chapter 1 (page 5). The main difference is that by using the exponential function, the calculations become shorter.

In a sense, the technical difficulties in dealing with the polynomial weight function from Claim 1.5 (page 19) are now hidden in the exponential function.

Remark: Technically, the real magic is happening inside Fact A.22 (page 770)—this is where the linear η term cancels out in the Taylor expansion and we're left with the much smaller η^2 term.

This is again analogous to what we observed more directly in the proof of Claim 1.5 (page 19). There, the cancellation took place after the application of the binomial theorem: the linear terms cancel, and only the even-order terms remain, dominated by the quadratic term.



We now present the second proof, which uses a symmetric weight function that naturally handles the zero-discrepancy case. We will denote the new weight function by $\omega^t(S)$, to avoid confusion with the previous weight function $y^t(S)$.

Proof of Theorem 1.4 (page 16) via a 'different weight function'. We use the following weight function, denoted by $\omega(\cdot)$:

$$\omega^t(S) = \frac{\exp\left(\eta \cdot \operatorname{disc}^t(S)\right) + \exp\left(-\eta \cdot \operatorname{disc}^t(S)\right)}{2}. \tag{2.5}$$

Set

$$\Omega^t = \sum_{S \in \mathcal{R}} \omega^t(S).$$

Note that for all $S \in \mathcal{R}$,

$$\operatorname{disc}^1(S) = 0 \qquad \text{and thus} \quad \omega^1(S) = 1 \quad \text{and} \quad \Omega^1 = m.$$

Then here is the corresponding algorithm, where $\eta > 0$ will be set optimally later.

Exp Discrepancy Algorithm 2 $\left(X = \{v_1, \ldots, v_n\}, \mathcal{R}\right)$

For $t = 1, \ldots, n$:

1. Assign v_t a color that realizes the following minimum:

$$\min\left\{ \sum_{S \in \mathcal{R}} \omega^{t+1}\left(S \mid v_t \text{ is given color '+1'}\right), \sum_{S \in \mathcal{R}} \omega^{t+1}\left(S \mid v_t \text{ is given color '-1'}\right) \right\}.$$

   If both colors give the same total weight, choose either one arbitrarily.

2. Update the weights: for each $S \in \mathcal{R}$, set

$$\omega^{t+1}(S) = \frac{\exp\left(\eta \cdot \operatorname{disc}^{t+1}(S)\right) + \exp\left(-\eta \cdot \operatorname{disc}^{t+1}(S)\right)}{2}.$$

We give the formal claim to bound the weight increase in a single iteration.

Claim 2.6. *At the start of the t-th iteration, we can assign a color to v_t such that*

$$\Omega^{t+1} = \sum_{S \in \mathcal{R}} \omega^{t+1}(S) \le \left(\frac{e^{\eta} + e^{-\eta}}{2}\right) \cdot \Omega^t.$$

Proof. For the sets that do not contain v_t, the weight remains unchanged. Fix a set $S \in \mathcal{R}$ that contains v_t. We consider two cases:

Case 1: $\operatorname{disc}^t(S) = 0$**.** In this case, irrespective of the color given to v_t, we have

$$\operatorname{disc}^{t+1}(S) = 1 \qquad \Longrightarrow \qquad \omega^{t+1}(S) = \left(\frac{e^{\eta} + e^{-\eta}}{2}\right) = \omega^t(S) \cdot \left(\frac{e^{\eta} + e^{-\eta}}{2}\right).$$

Case 2: $\operatorname{disc}^t(S) > 0$**.** Here, one of the color assignments for v_t must increase its discrepancy, and the other must decrease it. Thus, we can compute the average weight of S after coloring v_t, over the two possible color choices.

$$\frac{1}{2}\left(\omega^{t+1}\left(S \mid v_t \text{ is given color '-1'}\right) + \omega^{t+1}\left(S \mid v_t \text{ is given color '+1'}\right)\right)$$

$$= \frac{1}{2}\left(\frac{e^{\eta\cdot(\mathrm{disc}^t(S)+1)} + e^{-\eta\cdot(\mathrm{disc}^t(S)+1)}}{2}\right) + \frac{1}{2}\left(\frac{e^{\eta\cdot(\mathrm{disc}^t(S)-1)} + e^{-\eta\cdot(\mathrm{disc}^t(S)-1)}}{2}\right)$$

$$= \frac{e^{\eta\,\mathrm{disc}^t(S)}\cdot e^{\eta}}{4} + \frac{e^{-\eta\,\mathrm{disc}^t(S)}\cdot e^{-\eta}}{4} + \frac{e^{\eta\,\mathrm{disc}^t(S)}\cdot e^{-\eta}}{4} + \frac{e^{-\eta\,\mathrm{disc}^t(S)}\cdot e^{\eta}}{4}$$

$$= \frac{e^{\eta\,\mathrm{disc}^t(S)}}{2}\left(\frac{e^{\eta}+e^{-\eta}}{2}\right) + \frac{e^{-\eta\,\mathrm{disc}^t(S)}}{2}\left(\frac{e^{\eta}+e^{-\eta}}{2}\right)$$

$$= \left(\frac{e^{\eta\,\mathrm{disc}^t(S)} + e^{-\eta\,\mathrm{disc}^t(S)}}{2}\right)\cdot\left(\frac{e^{\eta}+e^{-\eta}}{2}\right) = \omega^t(S)\cdot\left(\frac{e^{\eta}+e^{-\eta}}{2}\right).$$

It follows that there is a color for v_t such that

$$\Omega^{t+1} \le \frac{1}{2}\left(\sum_{S\in\mathcal{R}}\omega^{t+1}\left(S \;\middle|\; v_t \text{ is given color '-1'}\right) + \sum_{S\in\mathcal{R}}\omega^{t+1}\left(S \;\middle|\; v_t \text{ is given color '+1'}\right)\right)$$

$$= \sum_{S\in\mathcal{R}}\frac{1}{2}\left(\omega^{t+1}\left(S \;\middle|\; v_t \text{ is given color '-1'}\right) + \omega^{t+1}\left(S \;\middle|\; v_t \text{ is given color '+1'}\right)\right)$$

$$\le \sum_{S\in\mathcal{R}}\omega^t(S)\cdot\left(\frac{e^{\eta}+e^{-\eta}}{2}\right) = \left(\frac{e^{\eta}+e^{-\eta}}{2}\right)\cdot\Omega^t.$$

□

We have $\Omega^1 = m$. Thus, upper and lower bounding the total weight after n iterations,

$$\max_{S\in\mathcal{R}}\omega^{n+1}(S) \;\le \boldsymbol{\Omega^{n+1}} \le\; m\cdot\left(\frac{e^{\eta}+e^{-\eta}}{2}\right)^n.$$

Using the inequality $e^{\eta}+e^{-\eta} \le 2e^{\eta^2/2}$ (Fact A.22 (page 770)) and substituting for $\omega^{n+1}(\cdot)$,

$$\max_{S\in\mathcal{R}}\frac{\exp\left(\eta\cdot\mathrm{disc}^{n+1}(S)\right) + \exp\left(-\eta\cdot\mathrm{disc}^{n+1}(S)\right)}{2} \;\le\; m\cdot e^{n\eta^2/2}.$$

Using $\exp(-\eta\,\mathrm{disc}^{n+1}(S)) > 0$,

$$\max_{S\in\mathcal{R}}\frac{\exp\left(\eta\cdot\mathrm{disc}^{n+1}(S)\right)}{2} \;\le\; m\cdot e^{n\eta^2/2}.$$

Taking logarithms and re-arranging terms,

$$\max_{S\in\mathcal{R}}\mathrm{disc}^{n+1}(S) \le \frac{\ln 2m}{\eta} + \frac{n\eta}{2}.$$

The above is minimized by setting $\eta = \Theta\left(\sqrt{\frac{\ln m}{n}}\right)$, giving the desired upper bound.

Remark: For the probabilistic version, set the color of v_t to '$+1$' or '-1' with equal probability. Then linearity of expectation gives

$$\mathrm{E}\left[\Omega^{t+1}\right] = \mathrm{E}\left[\sum_{S\in\mathcal{R}} \omega^{t+1}(S)\right] = \sum_{S\in\mathcal{R}} \mathrm{E}\left[\omega^{t+1}(S)\right] \leq \sum_{S\in\mathcal{R}} \left(\frac{e^{\eta}+e^{-\eta}}{2}\right) \omega^{t}(S) = \left(\frac{e^{\eta}+e^{-\eta}}{2}\right) \Omega^{t}.$$



As the key cancellation is happening inside Fact A.22 (page 770), we give its proof.

Proof. Using the Taylor series of $\exp$ at 0 gives, for any $\eta > 0$,

$$e^{\eta} = 1 + \frac{\eta}{1!} + \frac{\eta^2}{2!} + \frac{\eta^3}{3!} + \frac{\eta^4}{4!} + \cdots$$

$$e^{-\eta} = 1 - \frac{\eta}{1!} + \frac{\eta^2}{2!} - \frac{\eta^3}{3!} + \frac{\eta^4}{4!} + \cdots$$

Adding them up, the linear terms cancel—so the quadratic term becomes the dominant one for $\eta < 1$—and we get

$$e^{\eta} + e^{-\eta} = 2\left(1 + \frac{\eta^2}{2!} + \frac{\eta^4}{4!} + \frac{\eta^6}{6!} + \frac{\eta^8}{8!} + \cdots\right)$$

Using the fact that $(2i)! = 2i \cdot (2i-1) \cdots (i+1) \cdot i! \geq 2^i i!$,

$$< 2\left(1 + \frac{\left(\eta^2\right)}{2^1 \cdot 1!} + \frac{\left(\eta^2\right)^2}{2^2 \cdot 2!} + \frac{\left(\eta^2\right)^3}{2^3 \cdot 3!} + \frac{\left(\eta^2\right)^4}{2^4 \cdot 4!} + \cdots\right)$$

$$= 2\left(1 + \frac{\left(\eta^2/2\right)}{1!} + \frac{\left(\eta^2/2\right)^2}{2!} + \frac{\left(\eta^2/2\right)^3}{3!} + \frac{\left(\eta^2/2\right)^4}{4!} + \cdots\right) = 2\,e^{\eta^2/2}.$$

□



Bibliography and discussion. The new weight function—that is, hyperbolic cosine $\left(e^{\eta}+e^{-\eta}\right)/2$—is from [OS78], which also makes the following prescient observation:

> It is interesting to note that our procedure $\cdots$ is *nonanticipative* in that we determine ϵ_i knowing only the first i rows of A (and the total number of rows m).

Another way one can arrive at the function $\frac{1}{2}\left(e^{\eta}+e^{-\eta}\right)$ is via the proof of the tail bound used to prove the $O\left(\sqrt{n \ln m}\right)$ bound for discrepancy via a random coloring; see [You95]. We refer the reader to the first chapter of Chazelle [Cha00] for a beautiful (though succinct) exposition.